

Lucas Narloch Zabla

Backend Developer | Python & FastAPI

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Backend Developer with production experience building Python and FastAPI applications, RESTful APIs, internal tools, and automation solutions in AWS environments. Experienced in integrating corporate systems, relational databases, event-driven services, and external platforms. Strong focus on clean and maintainable code, automated testing, logging, troubleshooting, infrastructure automation, and continuous improvement.

TECHNICAL SKILLS

* **Backend:** Python, FastAPI, Node.js, TypeScript, NestJS, RESTful APIs, Object-Oriented Programming * **Databases:** PostgreSQL, MySQL, DynamoDB, SQL * **Testing & Quality:** pytest, Testcontainers, unit testing, integration testing, error handling, logging, Clean Code * **Cloud & Infrastructure:** AWS Lambda, S3, SQS, EC2, RDS, IoT Core, CloudWatch, Terraform, Docker * **Automation & DevOps:** GitHub Actions, CI/CD, Bash, Linux, infrastructure as code, deployment automation * **Security:** JWT, OAuth 2.0, OpenID Connect, AWS Cognito, authentication and authorization * **Tools:** Git, GitHub, Linux, Docker, Nginx, Prometheus, Grafana

EXPERIENCE

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

09/2024 - Present

* Develop and maintain backend applications and RESTful APIs using Python and FastAPI in an AWS production environment. * Contributed to consolidating multiple serverless functions into a structured FastAPI application, centralizing business rules, reducing operational complexity, and improving maintainability. * Developed an internal endpoint for migrating data between IoT devices, integrating MySQL, DynamoDB, Amazon S3, and AWS IoT Core through MQTT and supporting thousands of historical records per operation. * Created automation solutions and asynchronous workflows with Amazon SQS, processing peaks above 40,000 notifications per day and removing blocking operations from the main request flow. * Implemented an automated test suite with pytest and Testcontainers, including 30 integration tests and 80 unit tests, increasing delivery reliability and reducing production regression risks. * Investigated production errors through structured logs, metrics, and AWS CloudWatch, optimizing Python Lambda functions and reducing the average number of errors every five minutes from 113 to 5, a 95.6% reduction. * Integrated backend services with relational databases, NoSQL databases, AWS services, IoT devices, and external authentication providers. * Worked with JWT, OAuth 2.0, OpenID Connect, and AWS Cognito to implement authentication, authorization, and account integration flows. * Automated application deployments and environment configuration with GitHub Actions, Terraform, Bash scripts, and Docker, reducing manual work and configuration risks. * Participated in technical discussions involving API design, system architecture, event-driven communication, observability, deployment, and production reliability. * Documented APIs, integrations, infrastructure workflows, and operational procedures to support system maintenance and team collaboration. * Collaborated with backend, frontend, firmware, and product teams to understand requirements and translate business needs into reliable technical solutions.

PROJECTS

Sunlog.dev

Production web platform designed for developers to document projects, technical decisions, and development progress. Built with a RESTful backend, PostgreSQL, automated infrastructure, authentication, background processing, and production observability.

The platform includes 33 HTTP endpoints, more than 100 automated tests, JWT-based authentication, asynchronous processing, CI/CD pipelines with GitHub Actions, Docker containerization, reverse proxy configuration, structured logging, and monitoring with Prometheus and Grafana.

Technologies: NestJS, TypeScript, PostgreSQL, JWT, Docker, Nginx, AWS, GitHub Actions, Prometheus, Grafana, Next.js

[Code](#) | [Live demo](#)

PocScript

Programming language and compiler implemented in C, covering the complete compilation pipeline, including lexical analysis, AST parsing, semantic analysis, custom intermediate representation generation, LLVM IR emission, and native executable generation.

Includes a custom runtime based on Linux system calls, a static type system, scope resolution, arrays, pointers, error handling, and 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux system calls, Compiler Design

[Code](#)

EDUCATION

Bachelor of Computer Science - PUCPR

02/2025 - 12/2028

Full scholarship recipient. Coursework focused on algorithms, data structures, object-oriented programming, operating systems, computer networks, databases, and software engineering.

Technical High School Diploma in Electronics - CEEP

02/2022 - 12/2024

Technical education focused on electronics, embedded systems, digital systems, automation, and programming.

CERTIFICATIONS

* **Fundamentals of Database Engineering** - Hussein Nasser, Udemy | Sep/2025

LANGUAGES

* **Portuguese:** Native * **English:** Professional Working Proficiency